

NEPHROLOGY TEACHING SERIES

# Hyponatremia

*Tonicity, volume status, and the art of slow correction*

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Chronic hyponatremia is a water problem. Fix it slowly, or you'll break the brain.

Dr. Cheng • Premier Nephrology Medical Group

*MS3/MS4 rotation teaching*

# Why we get consulted

*Diagnosis looks simple; overcorrection isn't.*

## TRIGGERS FOR CONSULT

- Na < 125 or drop  $\geq 10$  in 24 h
- Symptomatic (seizure, AMS)
- Unclear etiology
- SIADH refractory to fluid restriction
- Cirrhosis / HF with progressive Na drop
- Overcorrection / DDAVP clamp for ODS

## TEACHING PEARL

*Three questions answer 90%: is it truly hypotonic? How concentrated is the urine? What's the volume status? Everything flows from these.*

# Three labs decide the diagnosis

*Order all three on the same draw. Don't skip serum osmolality.*

## SERUM OSMOLALITY

Normal: 275–295 mOsm/kg

< 275 → TRUE hypotonic hyponatremia — proceed

275–295 → isotonic (pseudohyponatremia from lipids/protein)

> 295 → hypertonic (hyperglycemia, mannitol)

## URINE OSMOLALITY

< 100 → water excess: primary polydipsia, beer potomania, tea & toast

100–300 → mixed / resetting

> 300 → ADH active: SIADH, hypovolemia, HF, cirrhosis, hypothyroid, adrenal insufficiency

## URINE SODIUM

< 20 → low effective circulating volume (true hypovolemia or HF/cirrhosis)

> 30 → SIADH or renal salt loss (diuretics, CSW, adrenal insufficiency)

Helpful only after urine osm shows ADH is on

# Etiology by volume status

*Once true hypotonic, volume status sorts the differential.*

## **HYPOVOLEMIC (true $\text{Na}^+$ loss > water loss)**

- GI losses, diuretics (thiazides top offender), bleeding • Cerebral salt wasting (neuro patients) • Adrenal insufficiency / mineralocorticoid deficiency
- Urine  $\text{Na} < 20$  (extrarenal);  $> 30$  (renal) • Treat with isotonic saline → stop the ADH drive

## **EUVOLEMIC (water excess, $\text{Na}^+$ normal)**

- SIADH (CNS, pulmonary, meds, malignancy) • Hypothyroidism • Glucocorticoid deficiency • Low solute intake (tea & toast, beer potomania) • Primary polydipsia • Management differs: fluid restriction  $\pm$  salt tabs  $\pm$  vaptan

## **HYPERVOLEMIC (impaired free water excretion)**

- Heart failure, cirrhosis, nephrotic syndrome, advanced CKD • All share low effective arterial blood volume → ADH on despite expansion • Fluid +  $\text{Na}^+$  restriction; treat underlying disease • Tolvaptan in hospitalized HF / cirrhosis with caution

# SIADH — the diagnosis of pattern recognition

*A clinical diagnosis that requires every exclusion before you commit.*

## DIAGNOSTIC CRITERIA (Bartter & Schwartz)

Hypotonic (serum osm  $< 275$ )

Urine osm  $> 100$

Clinically euvolemic

Urine Na  $> 30$

Normal TSH + cortisol

No recent diuretic

Responds to fluid restriction

## CAUSES — 4 BUCKETS

**CNS:** stroke, SAH, trauma, meningitis

**Pulmonary:** pneumonia, TB, small-cell lung

**Meds:** SSRIs, carbamazepine, MDMA, thiazides

**Malignancy** (ectopic ADH)

Transient: post-op, pain, nausea

# Correction rate — the central safety question

*Goal: rise the Na by the right amount in the right time frame — no more.*

## ACUTE ( < 48 h or severe symptoms )

Seizure, coma, respiratory arrest = brain swelling

**3% saline 100 mL bolus IV over 10 min (may repeat ×2 until symptoms resolve)**

Goal: raise  $\text{Na}^+$  by 4–6 mEq/L quickly

Then cap at  $\leq 8$  mEq/L in 24 h unless clearly acute (< 48 h)

Use 3% infusion 0.5–2 mL/kg/h after bolus

## CHRONIC ( > 48 h or unknown duration )

Brain has adapted — correcting fast causes ODS

**Cap:  $\leq 8$  mEq/L per 24 h for everyone (aim 4–6 if high risk: alcoholism, malnutrition, hypoK)**

Check  $\text{Na}^+$  every 2 h initially

If overshooting: DDAVP 2–4  $\mu\text{g}$  IV q6h + D5W

'DDAVP clamp' — pre-empt overcorrection in high-risk patients

# Treatment — match the tool to the phenotype

Volume status drives the choice as much as the  $\text{Na}^+$  value.

| Phenotype                            | First-line                                       | Second-line                                                                                                     | Avoid                                                                     |
|--------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Severe symptomatic (acute)</b>    | 3% saline 100 mL bolus × up to 3                 | Continuous 3% at 0.5–2 mL/kg/h; re-check q2h                                                                    | <i>Isotonic saline (may worsen if SIADH)</i>                              |
| <b>SIADH (euvolemic)</b>             | Fluid restriction 800–1000 mL/day                | Salt tabs 1–2 g TID ± loop diuretic; selected inpatient tolvaptan if refractory                                 | <i>Isotonic saline (desalination can worsen <math>\text{Na}^+</math>)</i> |
| <b>Hypovolemic</b>                   | Isotonic saline cautiously                       | Monitor for brisk overcorrection once volume restored                                                           | <i>Fluid restriction (worsens ADH drive)</i>                              |
| <b>Hypervolemic (HF/cirrhosis)</b>   | Fluid + $\text{Na}^+$ restriction; loop diuretic | HF: selected inpatient tolvaptan.<br>Cirrhosis/liver disease: generally avoid or use only with hepatology input | <i>Large volumes of saline; vaptans in liver failure</i>                  |
| <b>Drug-induced (thiazide, SSRI)</b> | Stop the drug                                    | Fluid restriction until Na normalizes; expect brisk rise → DDAVP if overcorrecting                              | <i>Resuming thiazide without reassessment</i>                             |

# Osmotic demyelination syndrome (ODS)

*The preventable complication that defines the field.*

## HIGH-RISK PATIENTS

- Baseline  $\text{Na}^+$  < 120
- Chronic hyponatremia (unknown duration = assume chronic)
- Alcohol use disorder
- Malnutrition / cachexia
- Hypokalemia ( $\text{K}^+$  repletion will drive  $\text{Na}^+$  up further)
- Liver disease / advanced cirrhosis
- Hypoxia

## PREVENTION / RESCUE

**Prevention:** limit  $\text{Na}^+$  rise to  $\leq 8 \text{ mEq/L/24 h}$  (aim 4–6 in high risk)

Check  $\text{Na}^+$  q2h during active correction

Replete  $\text{K}^+$  into the Na budget (K rises raise Na too)

**Rescue: if overshooting, DDAVP 2–4  $\mu\text{g}$  IV + D5W 3 mL/kg over 1 h**

Goal: re-lower  $\text{Na}^+$  to stay under threshold

Clinical ODS may not appear for 2–6 days — act on labs, not symptoms

# Common pitfalls

*These are the ones that lead to ODS or to missed diagnoses.*

## AVOID

- Skipping serum osmolality
- Calling SIADH without TSH / cortisol
- Isotonic saline in true SIADH
- Fluid restriction in hypovolemia
- Not checking K (K raises Na)
- Correcting without Na q2h

## DO INSTEAD

- Always order: serum osm, urine osm, urine Na
- Rule out thyroid / adrenal first
- SIADH → fluid restriction, not saline
- Hypovolemia → saline, watch rebound
- Budget K repletion into 24-h  $\Delta$ Na
- q2h labs until stable

# Clinical case

*Think it through — answer is on the next slide.*

## VIGNETTE

A 74-year-old woman with small-cell lung cancer presents with 3 days of progressive confusion. Na<sup>+</sup> 114 (baseline 138 one month ago), no seizures. Exam: alert but disoriented × 2. Euvolemic. Serum osm 244, urine osm 560, urine Na 48. TSH and AM cortisol normal. On home sertraline; no diuretic. On admission she's given 500 mL NS.

**QUESTION:** What's the diagnosis, is this acute or chronic, and what's the correction target?

# Clinical case — answer

*Key teaching points from the case.*

## ANSWER

**SIADH secondary to small-cell lung cancer; chronic hyponatremia (> 48 h); cap the Na<sup>+</sup> rise at ≤ 8 mEq/L per 24 h (aim 4–6) to avoid ODS.**

## TEACHING

Classic SIADH: hypotonic hyponatremia, inappropriately concentrated urine, urine Na > 30, euvolemic, normal TSH/cortisol. Isotonic saline can worsen SIADH via desalination. Treatment: fluid restriction (800-1000 mL/day) ± salt tabs; consider inpatient tolvaptan only if refractory and closely monitored. Na<sup>+</sup> q2h during correction. If overcorrecting, rescue with DDAVP 2-4 µg IV + D5W 3 mL/kg. High-risk for ODS: alcoholism, malnutrition, hypokalemia, Na<sup>+</sup> < 120.

## REFERENCES

# Hyponatremia

### Guidelines

Expert panel consensus (Am J Med 2013) on hyponatremia  
European clinical practice guideline — ERA-EDTA / ESE 2014  
Endocrine Society — SIADH guidance  
Hoorn & Zietse. JASN 2017 — practical correction approach

### Landmark Trials

SALT-1 / SALT-2 — tolvaptan in SIADH, HF, cirrhosis  
SALTWATER — long-term tolvaptan safety  
Sterns et al. — ODS risk and correction pacing  
Garrahy et al. — empiric fluid restriction success rates

### UpToDate (2026 access):

*Causes of hyponatremia in adults • Overview of the treatment of hyponatremia in adults • Treatment of hyponatremia: SIADH and reset osmostat • Osmotic demyelination syndrome and overly rapid correction*